

Docker Lab 07 - Port Mapping and Troubleshooting

Objective

Learn how Docker port mapping works, access containers from a browser, inspect networking information, and troubleshoot port and container name conflicts.

Commands Used

Create Nginx Container

```
docker run -d --name web -p 8080:80 nginx
```

Purpose:

- Create Nginx container
 - Run in detached mode
 - Map host port 8080 to container port 80
-

Verify Running Containers

```
docker ps
```

Purpose:

- Show running containers
 - Verify port mappings
 - Confirm container status
-

Show All Containers

```
docker ps -a
```

Purpose:

- Display running and stopped containers

Inspect Container Information

```
docker inspect web
```

Purpose:

- View detailed container configuration
- Networking information
- Status and metadata

Find Container Image

```
docker inspect web | findstr Image
```

Purpose:

- Identify image used by the container

Find Container IP Address

```
docker inspect web | findstr IPAddress
```

Result:

```
172.17.0.2
```

Purpose:

- View Docker internal IP address

Find Container Status

```
docker inspect web | findstr Status
```

Purpose:

- Verify running state
-

View Port Mapping

```
docker port web
```

Result:

```
80/tcp -> 0.0.0.0:8080  
80/tcp -> [::]:8080
```

Purpose:

- Display host-to-container port mapping
-

View Running Processes

```
docker top web
```

Purpose:

- View processes running inside container

Result:

```
nginx master process  
nginx worker process
```

View Container Logs

```
docker logs web
```

Purpose:

- Inspect container startup logs
 - Verify Nginx service startup
-

View Resource Usage

```
docker stats
```

Purpose:

- Monitor CPU usage
 - Monitor memory usage
 - Monitor network traffic
 - Monitor running containers
-

Browser Testing

Open:

```
http://localhost:8080
```

Result:

```
Welcome to nginx!
```

Purpose:

- Verify Nginx is accessible from host machine
-

Troubleshooting Performed

Error 1 - Missing Image Name

Command:

```
docker run -d --name web2 -p8080:80
```

Error:

```
docker run requires at least 1 argument
```

Cause:

- Image name was missing

Fix:

```
docker run -d --name web2 -p 8080:80 nginx
```

Error 2 - Port Conflict

Command:

```
docker run -d --name web2 -p 8080:80 nginx
```

Error:

```
Bind for 0.0.0.0:8080 failed:  
port is already allocated
```

Cause:

- Container "web" was already using host port 8080

Fix:

- Use a different host port
- Or stop existing container

Example:

```
docker run -d --name web2 -p 9090:80 nginx
```

Error 3 - Container Name Conflict

Error:

```
container name "/web2" is already in use
```

Cause:

- Failed container was already created

Fix:

```
docker rm web2
```

Then recreate:

```
docker run -d --name web2 -p 9090:80 nginx
```

Successful Deployment

```
docker run -d --name web2 -p 9090:80 nginx
```

Verify:

```
docker ps
```

Result:

```
web    -> 8080:80
web2   -> 9090:80
```

Git Commands Used

Check Status

```
git status
```

Add Files

```
git add .
```

Commit Changes

```
git commit -m "Add Docker Lab 07 - Port Mapping and Troubleshooting"
```

Pull Remote Changes

```
git pull origin main
```

Push To GitHub

```
git push origin main
```

README Troubleshooting

Issue:

Images not displaying on GitHub.

Cause:

Markdown lines contained escape characters.

Incorrect:

```
\![Nginx Welcome](screenshots/01-nginx-browser.png)
```

Correct:

```
![Nginx Welcome](screenshots/01-nginx-browser.png)
```

Fix:

- Remove backslashes
- Save README
- Commit changes
- Push to GitHub

Concepts Learned

- Docker Port Mapping
- Host Port vs Container Port
- Docker Networking
- Container Inspection
- Docker Logs
- Docker Statistics
- Browser Access Testing
- Port Conflict Troubleshooting
- Container Name Conflict Troubleshooting
- Git Workflow
- GitHub Documentation
- Markdown Image Rendering

Key Takeaway

Port Mapping Example:

```
8080:80
```

Meaning:


```
Host Port      = 8080  
Container Port = 80
```

A host port can only be used by one service at a time.

Containers may use the same internal port as long as different host ports are assigned.